Scan Report: Scanning & Enumeration

Course: NIAI NAVTTC — Certified Ethical Hacker (CEH) Course

Assignment: Week 3 Lab Task

Prepared By: Muhammad Usman Saleem

Important Notice: All tasks documented in this report were performed only in an isolated virtual lab environment. Never test on real networks or systems without explicit written permission.

1. Executive Summary

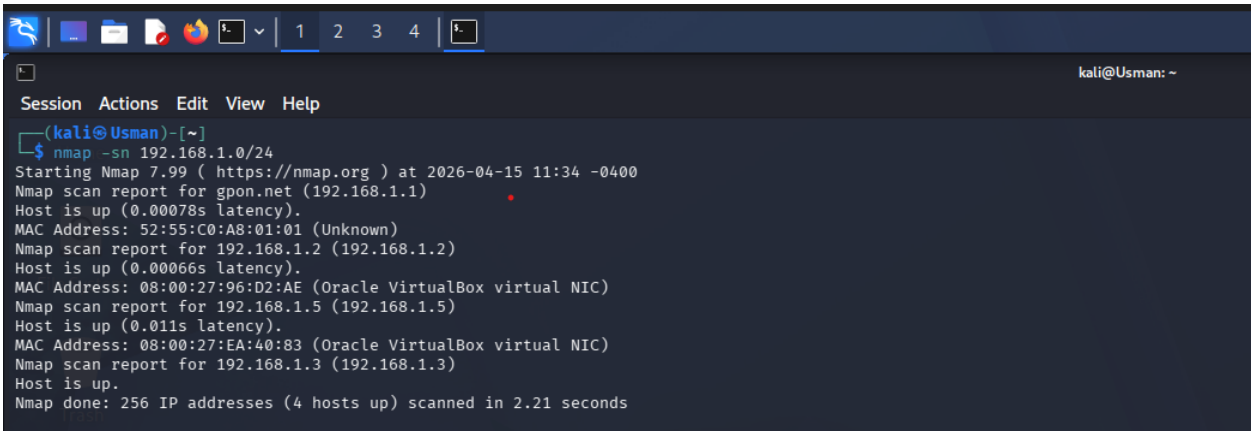
- **Objective:** Conduct a comprehensive network scan to identify all active hosts and services, and perform vulnerability scanning on identified hosts. Perform active enumeration to identify users, groups, and systems.
- **Target Machine:** Metasploitable 2
- **Target IP Address:** 192.168.1.5

---

2. Host Discovery (Ping Sweep)

Command used: nmap -sn 192.168.1.0/24

- **Active Hosts Found:** 5 total hosts up on the network.
    - 192.168.1.1 (Hostname: gpon.net) - MAC: QEMU virtual NIC
    - 192.168.1.2 - MAC: QEMU virtual NIC
    - 192.168.1.3 - MAC: Oracle VirtualBox virtual NIC
    - 192.168.1.4 (Kali Linux Host)
    - 192.168.1.5 (Metasploitable 2 Target) - MAC: Oracle VirtualBox virtual NIC



---

3. Port Scanning & Service Detection

Command used: nmap -sV -p- 192.168.1.5

- **Open Ports & Services:**

Port/Protocol	State	Service	Version
21/tcp	open	ftp	vsftpd 2.3.4
22/tcp	open	ssh	OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp	open	telnet	Linux telnetd
25/tcp	open	smtp	Postfix smtpd
53/tcp	open	domain	ISC BIND 9.4.2
80/tcp	open	http	Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp	open	rpcbind	2 (RPC #100000)
139/tcp	open	netbios-ssn	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp	open	netbios-ssn	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp	open	exec?	
513/tcp	open	login	

Port/Protocol	State	Service	Version
514/tcp	open	tcpwrapped	
1099/tcp	open	java-rmi	GNU Classpath grmiregistry
1524/tcp	open	bindshell	Metasploitable root shell
2049/tcp	open	nfs	2-4 (RPC #100003)
2121/tcp	open	ftp	ProFTPD 1.3.1
3306/tcp	open	mysql	MySQL 5.0.51a-3ubuntu5
3632/tcp	open	distccd	distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp	open	postgresql	PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp	open	vnc	VNC (protocol 3.3)
6000/tcp	open	X11	(access denied)
6667/tcp	open	irc	UnrealIRCd
6697/tcp	open	irc	UnrealIRCd
8180/tcp	open	unknown	
8787/tcp	open	drb	Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drb)
36043/tcp	open	nlockmgr	1-4 (RPC #100021)
53262/tcp	open	java-rmi	GNU Classpath grmiregistry
54124/tcp	open	mountd	1-3 (RPC #100005)
58160/tcp	open	status	1 (RPC #100024)

- **Additional Service Info:** Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux.

```
┌──(kali㉿Usman)-[~]
└─$ nmap -sV -p- 192.168.1.5
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-15 11:37 -0400
Stats: 0:00:55 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 93.33% done; ETC: 11:38 (0:00:02 remaining)
Nmap scan report for 192.168.1.5 (192.168.1.5)
Host is up (0.0022s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE     VERSION
21/tcp    open  ftp         vsftpd 2.3.4
22/tcp    open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet      Linux telnetd
25/tcp    open  smtp        Postfix smtpd
53/tcp    open  domain      ISC BIND 9.4.2
80/tcp    open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind     2 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec?
513/tcp   open  login       OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi    GNU Classpath grmiregistry
1524/tcp  open  bindshell   Metasploitable root shell
2049/tcp  open  nfs         2-4 (RPC #100003)
2121/tcp  open  ftp         ProFTPD 1.3.1
3306/tcp  open  mysql       MySQL 5.0.51a-3ubuntu5
3632/tcp  open  distccd     distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp  open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc         VNC (protocol 3.3)
6000/tcp  open  X11         (access denied)
6667/tcp  open  irc         UnrealIRCd
6697/tcp  open  irc         UnrealIRCd
8009/tcp  open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp  open  http        Apache Tomcat/Coyote JSP engine 1.1
8787/tcp  open  drb         Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drb)
36960/tcp open  mountd      1-3 (RPC #100005)
38268/tcp open  status      1 (RPC #100024)
58569/tcp open  java-rmi    GNU Classpath grmiregistry
59322/tcp open  nlockmgr    1-4 (RPC #100021)
MAC Address: 08:00:27:EA:40:83 (Oracle VirtualBox virtual NIC)
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 152.16 seconds
```

---

4. Operating System Detection

Command used: nmap -O 192.168.1.5

- **Operating System Detected:** Linux 2.6.X
- **OS Details:** Linux kernel 2.6.9 - 2.6.33
- **Device Type:** General purpose
- **Network Distance:** 1 hop

```
┌──(kali㉿Usman)-[~]
└─$ nmap -O 192.168.1.5
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-15 11:41 -0400
Nmap scan report for 192.168.1.5 (192.168.1.5)
Host is up (0.0095s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:EA:40:83 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.14 seconds
```

---

5. Vulnerability Scanning

Command used: nmap --script vuln 192.168.1.5

Vulnerabilities Flagged:

- **Port 21 (FTP):** vsFTPd version 2.3.4 backdoor (CVE-2011-2523). *Highly critical: allows immediate root shell access.*
- **Port 25 (SMTP):** SSL POODLE information leak (CVE-2014-3566), Anonymous Diffie-Hellman Key Exchange MitM Vulnerability, and TLS Protocol DHE_EXPORT Ciphers Downgrade MitM (Logjam - CVE-2015-4000).
- **Port 80 (HTTP):** Multiple potential SQL Injection vulnerabilities (e.g., within Mutillidae pages) and potential CSRF vulnerabilities identified. Discovered exposed directories (/phpMyAdmin/, /doc/, /icons/).
- **Port 1099 (rmiregistry):** RMI registry default configuration allows remote code execution (RCE).
- **Port 5432 (PostgreSQL):** SSL/TLS MITM vulnerability (CCS Injection - CVE-2014-0224) and SSL POODLE information leak.
- **Port 6667 (IRC):** Flagged as a trojaned version of UnrealIRCd containing a known backdoor.

```
┌──(kali㉿Usman)-[~]
└─$ nmap --script vuln 192.168.1.5
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-15 11:43 -0400
Nmap scan report for 192.168.1.5 (192.168.1.5)
Host is up (0.0011s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
| ftp-vsftpd-backdoor:
|   VULNERABLE:
|   vsFTPd version 2.3.4 backdoor
|     State: VULNERABLE (Exploitable)
|     IDs:  BID:48539  CVE:CVE-2011-2523
|       vsFTPd version 2.3.4 backdoor, this was reported on 2011-07-04.
|     Disclosure date: 2011-07-03
|     Exploit results:
|       Shell command: id
|       Results: uid=0(root) gid=0(root)
|     References:
|       https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb
|       http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html
|       https://www.securityfocus.com/bid/48539
|_      https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
|_sslv2-drown: ERROR: Script execution failed (use -d to debug)
| smtp-vuln-cve2010-4344:
|_  The SMTP server is not Exim: NOT VULNERABLE
| ssl-poodle:
|   VULNERABLE:
|   SSL POODLE information leak
|     State: VULNERABLE
|     IDs:  BID:70574  CVE:CVE-2014-3566
|           The SSL protocol 3.0, as used in OpenSSL through 1.0.1i and other
|           products, uses nondeterministic CBC padding, which makes it easier
|           for man-in-the-middle attackers to obtain cleartext data via a
|           padding-oracle attack, aka the "POODLE" issue.
|     Disclosure date: 2014-10-14
|     Check results:
|       TLS_RSA_WITH_AES_128_CBC_SHA
|     References:
|       https://www.openssl.org/~bodo/ssl-poodle.pdf
|       https://www.imperialviolet.org/2014/10/14/poodle.html
|       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2014-3566
|_      https://www.securityfocus.com/bid/70574
| ssl-dh-params:
|   VULNERABLE:
|   Anonymous Diffie-Hellman Key Exchange MitM Vulnerability
|     State: VULNERABLE
```

---

6. Active Enumeration

Commands used: nmap --script smb-enum-users 192.168.1.5

- nmap --script smb-enum-shares 192.168.1.5
- **Users Found:**
  - **Active/Normal Accounts:** msfadmin (RID: 3000), user (RID: 3002)
  - **Disabled/System Accounts:** Multiple built-in and service accounts were identified, including root, backup, bin, daemon, ftp, mysql, postgres, www-data, and others.
- **Groups Found:** *(Not explicitly queried in this SMB script run, but implied via standard Linux/Samba defaults)*
- **Shares Found:**
  - \\192.168.1.5\IPC$ - IPC Service (Anonymous access: **READ/WRITE**)
  - \\192.168.1.5\tmp - Type: DISKTREE, Comment: "oh noes!" (Anonymous access: **READ/WRITE**) - *High Risk*
  - \\192.168.1.5\ADMIN$ - IPC Service (Anonymous access: None)
  - \\192.168.1.5\opt - Type: DISKTREE (Anonymous access: None)

- \\192.168.1.5\print$ - Printer Drivers (Anonymous access: None)

```
┌──(kali㉿Usman)-[~]
└─$ nmap --script smb-enum-users 192.168.1.5
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-15 11:53 -0400
Nmap scan report for 192.168.1.5 (192.168.1.5)
Host is up (0.018s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:EA:40:83 (Oracle VirtualBox virtual NIC)
```

```
┌──(kali㉿Usman)-[~]
└─$ nmap --script smb-enum-shares  192.168.1.5
'Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-15 11:55 -0400
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 1 undergoing ARP Ping Scan
ARP Ping Scan Timing: About 100.00% done; ETC: 11:55 (0:00:00 remaining)
Nmap scan report for 192.168.1.5 (192.168.1.5)
Host is up (0.0016s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:EA:40:83 (Oracle VirtualBox virtual NIC)
```

---

7. Conclusion

- **Summary of Target: IP Address:** 192.168.1.5
    - **Hostname/System:** Metasploitable 2 (metasploitable.localdomain)
- **Final Assessment:** A comprehensive network and vulnerability scan was conducted against the target IP 192.168.1.5. The host was identified as running an outdated Linux 2.6.X kernel. The system presents a massive attack surface with 28 open TCP ports exposing various outdated services, including FTP, SSH, Telnet, HTTP, SMB, databases (MySQL/PostgreSQL), and IRC.

The vulnerability assessment flagged several critical, easily exploitable weaknesses:

- **Critical Backdoors:** The vsFTPd service (Port 21) and UnrealIRCd service (Port 6667) contain known backdoors that allow for immediate remote code execution and root access.
- **Web Application Flaws:** The HTTP service (Port 80) hosts vulnerable applications (like Mutillidae) susceptible to multiple SQL Injections and CSRF attacks.
- **Misconfigurations:** The Java RMI registry (Port 1099) allows for remote class loading, and older SSL/TLS configurations are vulnerable to POODLE and CCS Injection.
- **Exposed Enumeration Data:** Active SMB enumeration successfully extracted valid user accounts (e.g., msfadmin, user) and revealed misconfigured network shares. Specifically, the \tmp and IPC$ shares allow anonymous READ/WRITE access, providing a trivial entry point for payload uploads.

Overall Risk: The target system is critically vulnerable and can be compromised through multiple distinct attack vectors.